Candidate Seat no………….

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech (IT) Examination

December 2015

IT212 Computer Organization & Microprocessor

Date:07.12.2015, Monday **Time: 1:30 p.m. To 4:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 **Answer the following:** **[07]**

a. Which of the followings would correctly translate the pseudo instruction move $t1, $t0 into real MIPS commands? **[01]**

(A) add $t1, $t0, $0 (B) add $t1, $t0, $t0

(C) add $t0, $t1, $t1 (D) addi $t0, $t1, 0

b. What is true for MIPS pseudo instruction? **[01]**

(A) It is converted in exactly 2 instructions.

(B) It is converted in to 1 or more instruction.

(C)It is converted in to 2 or more instruction. (D)All of the above.

c. The Instruction used to move data from register to main memory is **[01]**

(A) lw (B) sw (C) li (D) both a & b (E) All of the above

d. Which of the following is used to keep track of starting base value of stack.? **[01]**

(A) $sp (B)$gp (C)$31 (D)$fp

e. The effectiveness of the cache memory is based on the property of ________. **[01]**

(A) Locality of reference (B) Memory localization

(C) Memory size (D) None of the above

f. What is the difference between logical address and physical address of a process. **[01]**

g. Give any two modes of I/O transfer. **[01]**

Q – 2.a What do you mean by Embedded Computers? Give at least 4 Examples of Embedded Computers. How are they different from general purpose computers? **[04]**

Q – 2.b What is a System Bus? Explain the different types of buses with function of each. **[05]**

OR

Q – 2.b Write MIPS code to multiply two numbers using ADD Instruction. **[05]**

Q – 2.c Multiply 14 and -5 using Booth's algorithm. **[05]**

OR

Q – 2.c What are Pipeline hazards? Explain any two in detail. **[05]**

Q - 3 **Answer the following:(Any Two)** **[14]**

a. Draw and explain single cycle data path for R-format Instructions.

b. How the rounding operation of floating point numbers is handled ?

c. How exceptions are handled in processor data path? Elaborate with example.

SECTION – II

Q - 4 **Do as directed:** **[07]**

a. An instruction pipeline can be implemented by means of **[01]**

(A) LIFO buffer (B) FIFO buffer (C) Stack (D) None of the above

b. The unit which decodes and translates each instruction and generates the necessary signals for ALU and other units is called **[01]**

(A) Arithmetic unit (B) Logical unit (C) Control unit (D) CPU

c. Logic gates with a set of input and outputs is arrangement of **[01]**

(A) Combinational circuit (B) Sequential circuit (C) Design circuits (D) Register

d. State True or False: Pipelining is a form of Implicit Parallelism. **[01]**

e. Convert -0.016510 in IEEE 754 single precision format. **[01]**

f. Give difference between sra and srl instruction with appropriate examples. **[02]**

Q – 5.a Consider the following code segment in C: **[04]**

A = B + E; C = B + F;

Here is the generated MIPS code for this segment, assuming all variables are in memory and are addressable as offsets from $t0:

```
lw $t1, 0($t0)
lw $t2, 4($t0)
add $t3, $t1,$t2
sw $t3, 12($t0)
lw $t4, 8($01)
add $t5, $t1,$t4
sw $t5, 16($t0)
```

Find the hazards in the above code segment and reorder the instructions to avoid any pipeline stalls.

Q – 5.b	What do you mean by Addressing Mode? Explain its types in detail with example.	**[05]**
	OR	
Q – 5.b	Draw and explain the flowchart for Instruction Cycle.	**[05]**
Q – 5.c	If clock rate is 50 MHZ, find execution time for a program with 1000 instructions , if CPI for program is 3.5? What if clock rate increases from 50MHZ to 250 MHZ and other functions remain same, how many times faster would computer be?	**[05]**
	OR	
Q – 5.c	What do you mean by Multilevel cache? Give its advantages.	**[05]**
Q – 6	**Attempt any Two:**	**[14]**
a.	Explain Direct mapped cache with example.	
b.	How does a Direct Memory Access Controller Work?	
c.	Write a short note on: RAID(Redundant Array of Independent Disk) Levels.	

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ALL THE BEST